Smith SS

AI Engineer | Agentic AI & LLM Specialist | Email: smithss.work@gmail.com | Phone: +65 88390915 | Portfolio: https://smith-s-s.github.io/portfolio/ | LinkedIn: www.linkedin.com/in/smith-s3 | GitHub: https://github.com/Smith-S-S | PR Resident, Singapore

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Summary

Highly accomplished **AI Engineer** with experience specializing in **Agentic AI systems**, **Multi-Agent Orchestration**, and **LLMOps**. Expert in designing **Retrieval-Augmented Generation (RAG)** pipelines, **MCP (Model Context Protocol)** servers, and autonomous robotic workflows. Proven track record in delivering scalable AI solutions that integrate **Computer Vision**, **Robotics**, and **Generative AI** to drive significant business impact and operational efficiency.

Technical Skills

- **Agentic AI & LLMs**: Agentic Workflows (ReAct, Reflection, Planning, Multi-Agent Collaboration, Sequential Workflows, Human-in-the-Loop), Multi-Agent Orchestration, MCP (Model Context Protocol), RAG, Prompt Engineering & Optimization, LLM Fine-Tuning, LangChain, LlamaIndex, CrewAI, AutoGen, HuggingFace, Cognitive Architectures, Tool Calling, Function Calling, Memory Persistence.
- **AI/ML Frameworks**: PyTorch, TensorFlow, OpenCV, YOLO (v5/v8/v11), PointNext (3D Segmentation), Scikit-learn, Pandas, NumPy.
- **Infrastructure & DevOps**: Docker, AWS (EC2, S3), Azure, GitHub Actions (CI/CD), MLflow, DVC, FastAPI, Fast MCP, Snowflake, Vector Databases (Pinecone, Supabase, Chroma).
- **Programming & Tools**: Python, PostgreSQL, n8n, Streamlit, Git, Linux.
- **Evaluation**: LangSmith.

Professional Experience

AI Engineer | Augmentus, Singapore

Jan 2026 – Present
Description: Architecting and deploying advanced agent-based AI systems for intelligent robotic workflow automation.

Roles & Responsibilities:

- Architected and deployed **multi-agent systems** for intelligent robotic workflow automation, leveraging **ReAct** and **Planning** patterns to handle complex, non-linear tasks and enable **autonomous reasoning**.
- Developed and deployed specialized **MCP (Model Context Protocol)** servers to standardize **tool orchestration** and data access for autonomous agents, improving system interoperability by 40% and facilitating **function calling**.

Technical Skills: Multi-Agent Systems, MCP, Tool Orchestration, Autonomous Reasoning, ReAct, Planning, Function Calling.

R&D AI Engineer | Augmentus, Singapore

Feb 2025 – Dec 2025
Description: Conducted research and development on agent-based AI systems, focusing on scalable orchestration, tool usage, and API-driven automation.

Roles & Responsibilities:

- Designed and deployed an **agent-based RAG pipeline** for automated UI review and redesign, reducing manual review time by **30%** through **autonomous feedback loops** and **reflection patterns**.
- Implemented a Self-reflecting **prompt validator and enhancer system** using LLMs, improving response relevance and accuracy by **25%** through advanced **prompt engineering**.
- Conducted R&D on scalable **multi-agent orchestration** and **tool usage**, achieving a **20% improvement** in system adaptability to new robotic tasks via robust **API integration**.

Technical Skills: RAG, Prompt Optimization, Reflection Patterns, API Integration, Multi-Agent Orchestration, Tool Usage.

Perception Intern | Augmentus, Singapore

Jul 2024 – Jan 2025
Description: Conducted research and development in 3D computer vision, focusing on point cloud segmentation for perception systems.

Roles & Responsibilities:

- Trained and optimized **PointNext** architecture for 3D segmentation, achieving a state-of-the-art **97% mIoU**, demonstrating expertise in **3D computer vision** and **deep learning optimization** for **point cloud processing**.

Technical Skills: 3D Computer Vision, Deep Learning Optimization, Point Cloud Processing, PointNext.

Key Projects

Personal AI Assistant Ecosystem (n8n & Agentic Orchestration)

Description: Developed a modular, low-code agentic orchestration system to function as a personal AI assistant.
Roles & Responsibilities:

- Built a modular **agentic orchestration system** using **n8n**, coordinating specialized sub-agents for Email, Calendar, Research, and Fitness management, demonstrating **low-code agentic workflows**.
- Developed a **Personal Data RAG Agent** and integrated **Fitbit API** for real-time activity tracking and health insights, showcasing **API orchestration** and **personal RAG** capabilities.

Technical Skills: n8n, Agentic Orchestration, Low-code Workflows, API Integration, RAG, Fitbit API.
Project Link: *https://shorturl.at/x6GdL*

End-to-End Predictive Maintenance (MLOps)

Description: Designed and deployed a comprehensive predictive maintenance machine learning pipeline with automated MLOps practices.
Roles & Responsibilities:

- Designed an end-to-end predictive ML pipeline with ETL data preparation, **MLflow** for experiment tracking, and automated hyperparameter tuning, achieving **98.8% accuracy**.

- Deployed the model on **AWS EC2** using **Docker** and **GitHub Actions** for automated **CI/CD**, demonstrating robust **MLOps** practices and **predictive modeling** expertise.

Technical Skills: MLOps, AWS EC2, Docker, GitHub Actions, MLflow, Predictive Modeling, CI/CD.
Project Link: *https://shorturl.at/ciQtz*

EngageVision Pro (Retail Analytics System)

Description: Developed a computer vision-based retail engagement tracking system for real-time analytics and customer behavior insights.
Roles & Responsibilities:

- Developed a **computer vision-based** retail engagement tracker using **YOLOv8**, **OpenCV**, and **Snowflake** for **real-time video analytics** and data storage.
- Built an interactive **Streamlit** dashboard to visualize customer behavior insights for 20+ pilot retail shops, enhancing data accessibility and demonstrating **data visualization** skills.

Technical Skills: Computer Vision, YOLOv8, OpenCV, Snowflake, Streamlit, Real-time Video Analytics, Data Visualization.
Project Link: *https://shorturl.at/9BK79*

Education

Bachelor of Engineering (Mechatronics, Robotics Specialization) *SRM Institute of Science and Technology, India | CGPA: 8.78/10 | Sep 2020 – Jun 2024*

Certifications

- **Machine Learning Specialization** – Stanford / Coursera (Andrew Ng), 2025
- **Deep Learning** – Skill-Lync, 2023
- **ServiceNow Certified Application Developer**, 2023